# ClipSAT : Pearson Edexcel IGCSE Mathematics A 4MA1 Higher Tier

**Unit 6.01: Sine & Cosine Rules, 3D Trigonometry & Vector Proofs** Name: _____

**Q1.** The coordinate diagram shows the parabolic curve f_1x = x^2 - 16x + 80 in the domain of Sine/Cosine Rules, 3D Trigonometry & Vectors. By completing the square f_1x = x - h^2 + k , find the sum k + f_110 .

[ ] A: 44
[ ] B: 36
[ ] C: 20
[ ] D: 34

**Q2.** A mathematical model in Sine/Cosine Rules, 3D Trigonometry & Vectors is defined by the cubic function y_2x = 4x^3 - 8x^2 + 15x - 13 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 123
[ ] B: 60
[ ] C: 75
[ ] D: 35

**Q3.** In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by mathbf{u}_{3} = langle 8, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 16
[ ] B: 128
[ ] C: 256
[ ] D: 144

**Q4.** The linear growth profile shown in the diagram for Sine/Cosine Rules, 3D Trigonometry & Vectors follows an arithmetic progression with first term a_1 = 9 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 765
[ ] B: 93
[ ] C: 855
[ ] D: 1395

**Q5.** In Sine/Cosine Rules, 3D Trigonometry & Vectors, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 32x,mathrm{d}x .

[ ] A: 136
[ ] B: 112
[ ] C: 144
[ ] D: 57

**Q6.** In Sine/Cosine Rules, 3D Trigonometry & Vectors, consider the quadratic function f_6x = x^2 - 18x + 169 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_616 .

[ ] A: 234
[ ] B: 137
[ ] C: 225
[ ] D: 218

**Q7.** The graph illustrates the cubic model y_7x = 4x^3 - 9x^2 + 33x - 20 studied in Sine/Cosine Rules, 3D Trigonometry & Vectors. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 801

[ ] B: 624

[ ] C: 111

[ ] D: 657

**Q8.** In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by mathbf{u}_{8} = langle 9, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 167

[ ] B: 37

[ ] C: 130

[ ] D: 237